Data Blocks

- Data Blocks (DB) are **structured memory areas** that can be allocated by the user
- They are stored in the **user memory** and **numbered as DB1, DB2, etc**. Their introduction takes place via the **TIA Portal**
- Similar to input/output/merker, they are **stored as bytes**, but they can be grouped (and referred) in **much more complex structures**
- Examples:
  - DB10.DBB7 is the **byte 7 of DB10**
  - DB3.DBW22 is the **word 22** (that comprises bytes 22 and 23) **of DB3**
  - DB99.DBD0 is the **double word 0** (that comprises bytes 0, 1, 2 and 3) of **DB99**
  - DB101.DBX33.2 is the **bit #2 of byte 33 of DB101**
- Good programming practice:
  - use only **even numbers** for data words
  - Use only **multiples of four numbers** for double data words

Operations with Data Block Operands/1

- Load and Transfer operations with byte/word/double words of Data Blocks have the **same syntax** of those already seen for inputs/outputs/merkers
- Examples**:**

```
L DB10.DBW4
T DB22.DBW0
```

The value of DBW4 of DB10 is copied to ACCU1
The value of ACCU1 (DB10.DBW4) is copied to DBW0 of DB22

```
L DB11.DBD16
T DB29.DBD4
```

The value of DBD16 of DB11 is copied to ACCU1
The value of ACCU1 (DB11.DBD16) is copied to DBD4 of DB29

- Bit operations for Data Blocks have the **same syntax** of those already seen for inputs/outputs/merkers
- Example**:**

```
U DB3.DBX7.6
U M0.0
U DB99.DBX0.1
```

AND operation between:
Bit #6 of byte 7 of DB3
Merker M0.0
Bit #1 of byte 0 of DB99

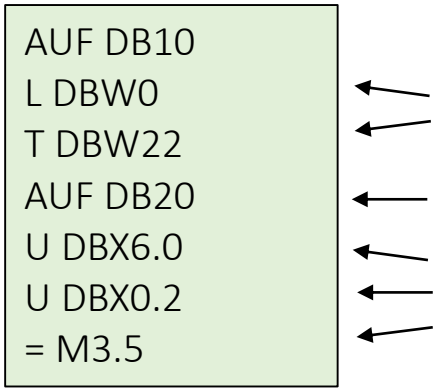
Operations with Data Block Operands/2

- If a sequence of operations involves the same Data Block, the number of the Data Block can be omitted
- To do this, the instruction **AUF DBn** can be used, that **opens DBn**. Example**:**



DB10 is open
The value of DBW0 of DB10 is copied to ACCU1
The value of ACCU1 (DB10.DBW0) is copied to DBW22 of DB10
Operation that assigns the value of bit 3 of byte 3 of DB10 to M2.1

- There is **not** a closing operation for DBs. To close a DB, it is necessary to open a new one. Example**:**



DB10 is open
Operations that involve Words of DB10

DB20 is Open and DB10 is closed

Operations that involves bits of DB20

Data Block Introduction

- A **dedicated TIA Portal tool** is available to introduce DBs
- As for the other blocks, Data Blocks have to be uploaded on the PLC
- DBs have a variable length with a maximum (1024 or 2048 bytes)
- If an instruction of the program refers to an operand of a DB **which is not present**, the **PLC enters the STOP state.**
- Example
  - Let's suppose we have uploaded a **DB with 15 bytes, from DBB0 to DBB14**
  - The instructions:
    - **L DBB20, O DBX30.4, T DBD12**… cause the PLC stop because they refer to operands that are not allocated
  - This represents a **severe programming error** because, in practice, we are trying to access memory areas which do not exist